

$$\therefore (0+b) = 0 + (b+b) \text{ (Algebraic property iv)}$$

$$\therefore b = (b+b) \text{ (Algebraic property iii)}$$

$$\Rightarrow (-b)+b = (-b)+(b+b) \Rightarrow (-b)+b = ((-b)+b)+b \text{ (Algebraic property 2)}$$

$$\Rightarrow 0 = 0+b \text{ (Algebraic property iv)} \Rightarrow b=0 \text{ (Algebraic property 3)}$$

Soln This is a contradiction.

$\therefore T(x+y) \neq T(x)+T(y)$. The first condition from the defn. of linear transformation is violated. $\square$

3. I und. 31. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let $\{v_1, v_2, v_3\}$ be a linearly dependent set in $\mathbb{R}^n$. Explain why the set $\{T(v_1), T(v_2), T(v_3)\}$ is linearly dependent.

Solution: Since $\{v_1, v_2, v_3\}$ are linearly dependent, by Theorem 7, there exists some vector which is a linear combination of v_1 and v_2 .

$\therefore$ WLOG, let v_3 be a linear combination of v_1 and v_2 .

$$\therefore v_3 = c_1 v_1 + c_2 v_2 \text{ for some } c_1, c_2 \in \mathbb{R}.$$

$$T(v_3) = T(c_1 v_1 + c_2 v_2) = T(c_1 v_1) + T(c_2 v_2) \text{ (by condition 1)}$$

$$= c_1 T(v_1) + c_2 T(v_2) \text{ (by condition 2)}$$

$T(v_3)$ is a linear combination of $T(v_1)$ and $T(v_2)$.

By Theorem 7, $\{T(v_1), T(v_2), T(v_3)\}$ is linearly dependent. $\square$

In Exercises 32-36, column vectors are written as rows, such as $x = (x_1, x_2)$, and $T(x)$ is written as $T(x_1, x_2)$.

32. Show that the transformation T defined by $T(x_1, x_2) = (4x_1 - 2x_2, 3|x_2|)$ is not linear.

$$\text{Solution: } T(x_1, x_2) = (4x_1 - 2x_2, 3|x_2|) \quad T(x_3, x_4) = (4x_3 - 2x_4, 3|x_4|)$$

$$T(x_1+x_3, x_2+x_4) = (4(x_1+x_3) - 2(x_2+x_4), 3|x_2+x_4|)$$

$$= (4x_1 - 2x_2 + 4x_3 - 2x_4, 3|x_2+x_4|)$$

$$= (4x_1 - 2x_2, 3|x_2|) + (4x_3 - 2x_4, 3|x_4|)$$

$$= (4x_1 - 2x_2, 3|x_2| + 3|x_4|)$$

(Vector addition)